

# Lecture 04 — Monte Carlo Simulation for Option Pricing

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

These notes cover Cluster 2: Monte Carlo (MC) integration, MC simulation of SDEs (Euler–Maruyama, Milstein), strong/weak convergence orders, and the four variance reduction techniques (antithetic variates, control variates, quasi-MC, importance sampling). The guiding theme: MC is the slow but dimension-robust benchmark against which the fast deterministic methods of this course — COS (Lectures 6, 10) and Carr–Madan FFT — are measured. MC converges at the model-independent rate  $O(N^{-1/2})$ ; COS converges *exponentially* in the number of terms for smooth densities. Knowing precisely why MC is  $O(N^{-1/2})$  is the way to articulate why COS is worth the trouble.

## Contents

|          |                                                                           |          |
|----------|---------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Monte Carlo integration and the <math>O(N^{-1/2})</math> error</b>     | <b>1</b> |
| 1.1      | The estimator . . . . .                                                   | 1        |
| 1.2      | Unbiasedness, consistency, and the standard error . . . . .               | 1        |
| 1.3      | Confidence intervals from the CLT . . . . .                               | 2        |
| <b>2</b> | <b>Pseudo-random vs. quasi-random (QMC)</b>                               | <b>2</b> |
| 2.1      | Why QMC can beat $O(N^{-1/2})$ . . . . .                                  | 2        |
| 2.2      | Caveats . . . . .                                                         | 3        |
| <b>3</b> | <b>Discretising SDEs</b>                                                  | <b>4</b> |
| 3.1      | Euler–Maruyama . . . . .                                                  | 4        |
| 3.2      | Milstein: deriving the extra term from the Itô–Taylor expansion . . . . . | 4        |
| <b>4</b> | <b>Strong vs. weak convergence</b>                                        | <b>5</b> |
| <b>5</b> | <b>Variance reduction</b>                                                 | <b>6</b> |
| 5.1      | Antithetic variates . . . . .                                             | 6        |
| 5.2      | Control variates: the optimal coefficient $c^*$ . . . . .                 | 6        |
| 5.3      | Importance sampling . . . . .                                             | 7        |
| <b>6</b> | <b>Where this sits in the course</b>                                      | <b>8</b> |
|          | <b>Honest gaps</b>                                                        | <b>8</b> |

## 1 Monte Carlo integration and the $O(N^{-1/2})$ error

### 1.1 The estimator

Let  $I = \int_a^b f(x) dx$  be a one-dimensional integral. Write it as an expectation by multiplying and dividing by the length  $b - a$ :

$$I = (b - a) \int_a^b f(x) \frac{1}{b - a} dx = (b - a) \mathbb{E}[f(U)], \quad U \sim \mathcal{U}[a, b],$$

since  $\frac{1}{b-a}\mathbf{1}_{[a,b]}$  is exactly the density of the uniform law on  $[a, b]$ . Drawing i.i.d.  $u_1, \dots, u_N \sim \mathcal{U}[a, b]$ , the estimator is

$$\hat{I}_N = \frac{b-a}{N} \sum_{i=1}^N f(u_i).$$

The general (multi-dimensional) form factorises the integrand  $f(\mathbf{x}) = h(\mathbf{x})g(\mathbf{x})$  with  $g$  a probability density on  $\Omega \subseteq \mathbb{R}^d$ , so that  $I = \int_{\Omega} h g d\mathbf{x} = \mathbb{E}_g[h(X)]$ , estimated by  $\hat{I}_N = \frac{1}{N} \sum_{i=1}^N h(x_i)$  with  $x_i \sim g$ . This factorisation is the whole game: *any* integral becomes an average of i.i.d. samples once you identify a density to sample from.

## 1.2 Unbiasedness, consistency, and the standard error

Let  $X_i = h(x_i)$ ,  $\mu = \mathbb{E}[h(X)] = I$ ,  $\sigma^2 = \text{Var}(h(X))$  (assume  $\sigma^2 < \infty$ ). The estimator  $\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i$  satisfies:

**Unbiasedness.**  $\mathbb{E}[\bar{X}_N] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[X_i] = \mu = I$ .

**Consistency.** The Strong Law of Large Numbers gives  $\bar{X}_N \xrightarrow{\text{a.s.}} \mu$ , so the estimator converges to the true integral for almost every realisation of the random stream. (The Weak Law gives the same in probability; a.s. is the stronger statement and is what the slides assert.)

**Variance of the estimator.** Using independence,

$$\text{Var}(\bar{X}_N) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(X_i) = \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}.$$

The cross terms vanish because  $\text{Cov}(X_i, X_j) = 0$  for  $i \neq j$ . Hence the *standard error* is

$$\text{se}(\bar{X}_N) = \sqrt{\text{Var}(\bar{X}_N)} = \frac{\sigma}{\sqrt{N}} = O(N^{-1/2}).$$

This is the central fact of the whole lecture: **the MC error decays like  $N^{-1/2}$ , independent of the dimension  $d$** . To gain one decimal digit (factor 10 in accuracy) you need  $100\times$  the samples. By contrast, a tensor-product quadrature rule of order  $k$  in  $d$  dimensions costs  $N = m^d$  points for  $m$  points per axis and achieves  $O(m^{-k}) = O(N^{-k/d})$  — the rate *degrades with  $d$*  (curse of dimensionality). MC's rate does not. This is why MC dominates in high dimension (basket options, many time steps) despite its slowness in low dimension.

## 1.3 Confidence intervals from the CLT

The Central Limit Theorem sharpens the standard error into a probabilistic error bar. With  $\mu = I$  and  $\sigma^2 = \text{Var}(X)$ ,

$$\frac{\bar{X}_N - I}{\sigma/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Therefore, for large  $N$ ,  $\bar{X}_N \approx \mathcal{N}(I, \sigma^2/N)$ , and an asymptotic  $(1 - \alpha)$  confidence interval for  $I$  is

$$\left[ \bar{X}_N - z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}}, \bar{X}_N + z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}} \right],$$

where  $z_{1-\alpha/2}$  is the standard normal quantile (e.g. 1.96 for 95%). In practice  $\sigma$  is unknown and is replaced by the sample standard deviation

$$s_N^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2,$$

(the  $N - 1$  gives an unbiased variance estimate). The half-width  $z_{1-\alpha/2} s_N / \sqrt{N}$  is the quantity actually reported alongside an MC price; a variance-reduction technique is "good" precisely when it shrinks this half-width for fixed  $N$  (equivalently, the same accuracy at fewer samples).

**Remark 1.** *The three convergence modes in the slides form a strict hierarchy:  $\xrightarrow{a.s.} \Rightarrow \xrightarrow{p} \Rightarrow \xrightarrow{d}$ , with no reverse implications in general. The SLLN is a.s., the WLLN is in probability, the CLT is in distribution. The CLT does not say  $\bar{X}_N \rightarrow I$  in distribution (that would be degenerate); it says the rescaled error  $\sqrt{N}(\bar{X}_N - I)/\sigma$  is asymptotically standard normal.*

## 2 Pseudo-random vs. quasi-random (QMC)

### 2.1 Why QMC can beat $O(N^{-1/2})$

Standard MC uses pseudo-random points that mimic i.i.d. uniforms; its error is *random* and decays at  $O(N^{-1/2})$  by the CLT. Quasi-Monte Carlo (QMC) replaces the random points by a *deterministic* low-discrepancy sequence  $u_1, \dots, u_N \in [0, 1]^d$  (Sobol, Halton, Faure, lattice rules) that fills the cube more evenly than random points, which clump and leave gaps. Both methods compute the *same* estimator  $\frac{1}{N} \sum_i f(x_i)$ ; they differ only in how the  $x_i$  are chosen.

The error is governed not by the CLT but by the deterministic **Koksma–Hlawka inequality**:

$$\left| \frac{1}{N} \sum_{i=1}^N f(x_i) - \int_{[0,1]^d} f(x) dx \right| \leq V(f) D_N^*,$$

a product of two factors that separate the integrand from the point set:

- $V(f)$  is the variation of  $f$  in the sense of Hardy–Krause, measuring the total magnitude of all mixed partial derivatives across coordinate subsets — how "rough"  $f$  is. It depends only on the integrand.
- $D_N^*$  is the *star discrepancy* of the point set  $P_N = \{x_1, \dots, x_N\}$ ,

$$D_N^*(P_N) = \sup_{t \in [0,1]^d} \left| \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{x_i \in [0,t]\}} - \prod_{j=1}^d t_j \right|,$$

the worst-case gap, over all origin-anchored boxes  $[0, t] = [0, t_1] \times \dots \times [0, t_d]$ , between the empirical fraction of points in the box and its Lebesgue volume  $\prod_j t_j$ . It depends only on the points.

For a low-discrepancy sequence one has

$$D_N^* = O\left(\frac{(\log N)^d}{N}\right),$$

so for sufficiently smooth (bounded-variation)  $f$  the QMC error is  $O((\log N)^d/N)$ . For fixed  $d$  this is *essentially*  $O(N^{-1})$  up to the logarithmic factor — a full power of  $N$  faster than the  $O(N^{-1/2})$  of plain MC.

### 2.2 Caveats

- The rate  $(\log N)^d/N$  carries the dimension in the exponent of the logarithm. For large  $d$ ,  $(\log N)^d$  is enormous and the asymptotic advantage only appears at impractically large  $N$ . In practice the *effective dimension* (how much each coordinate actually matters) is what governs QMC performance; Brownian-bridge or PCA constructions of the path lower the effective dimension and are what make QMC work for option pricing.

- QMC is biased-free but *deterministic*, so a plain QMC run gives no error estimate. *Randomised QMC* (e.g. random shifts of a lattice rule,  $x_n = \{nz/N + \Delta\} \bmod 1$  with random  $\Delta$ ) restores an unbiased estimator and lets you estimate the error from independent shifts, while keeping most of the discrepancy advantage.
- Mapping to non-unit / unbounded domains: scale  $[0, 1]^d$  linearly to  $[a, b]^d$  via  $x_i^{\text{new}} = a + (b - a)x_i$ , and map to a target law by the inverse CDF,  $x_i = \Phi^{-1}(u_i)$  (for Gaussians). The inverse-CDF transform preserves low-discrepancy structure better than Box–Muller.

**Remark 2** (Connection to COS/FFT). *QMC, like COS, exploits smoothness to beat  $O(N^{-1/2})$  — but QMC trades on smoothness/low variation of the integrand whereas COS trades on smoothness of the density, which it never touches directly, working instead with its Fourier-cosine coefficients (the characteristic function). When the characteristic function is known in closed form (Heston, Lévy models), COS achieves exponential convergence, far beyond even QMC’s near- $O(N^{-1})$ .*

### 3 Discretising SDEs

When the terminal law of  $X_T$  is not available in closed form, MC pricing requires simulating paths of the SDE

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t.$$

On a uniform grid  $t_k = kh$ ,  $k = 0, \dots, K$ ,  $Kh = T$ , with i.i.d.  $Z_k \sim \mathcal{N}(0, 1)$ , the increments  $W_{t_k} - W_{t_{k-1}} \stackrel{d}{=} \sqrt{h} Z_k$ .

#### 3.1 Euler–Maruyama

The simplest scheme freezes the coefficients at the left endpoint of each interval:

$$\hat{X}_{t_k} = \hat{X}_{t_{k-1}} + \mu(t_{k-1}, \hat{X}_{t_{k-1}}) h + \sigma(t_{k-1}, \hat{X}_{t_{k-1}}) \sqrt{h} Z_k, \quad \hat{X}_0 = X_0.$$

This is the Itô (left-endpoint) discretisation of  $X_{t_k} = X_{t_{k-1}} + \int_{t_{k-1}}^{t_k} \mu dt + \int_{t_{k-1}}^{t_k} \sigma dW$ , with each integral approximated by its left-point value times the increment. The left-endpoint choice is not cosmetic: it is what makes the stochastic integral a martingale (the Itô convention), as the  $\int W dW$  example in the slides shows.

#### 3.2 Milstein: deriving the extra term from the Itô–Taylor expansion

Euler ignores the variation of  $\sigma$  *within* a step. Milstein restores the leading correction. Work on one step  $[t_{k-1}, t_k]$ , write  $t_{k-1} = s$ , freeze  $X_s = x$ , and consider (for clarity) the autonomous scalar case  $dX_t = \mu(X_t) dt + \sigma(X_t) dW_t$ . The exact increment is

$$X_{t_k} - X_s = \int_s^{t_k} \mu(X_u) du + \int_s^{t_k} \sigma(X_u) dW_u.$$

Euler replaces  $\mu(X_u) \approx \mu(x)$  and  $\sigma(X_u) \approx \sigma(x)$ . The  $dt$  integral is already  $O(h)$  and its error is higher order, so the gain must come from the diffusion integral. Expand  $\sigma(X_u)$  around  $X_s = x$  by Itô’s lemma applied to the function  $\sigma$ :

$$d\sigma(X_t) = \sigma'(X_t) dX_t + \frac{1}{2} \sigma''(X_t) (dX_t)^2 = \left[ \sigma' \mu + \frac{1}{2} \sigma'' \sigma^2 \right] dt + \sigma' \sigma dW_t,$$

using the Itô multiplication rule  $(dW_t)^2 = dt$ . Integrating from  $s$  to  $u$ ,

$$\sigma(X_u) = \sigma(x) + \int_s^u \sigma'(X_r) \sigma(X_r) dW_r + (O(dt) \text{ terms}).$$

Substitute the *leading* stochastic correction back into the diffusion integral and freeze the inner coefficient at  $x$  (the rest is higher order):

$$\int_s^{t_k} \sigma(X_u) dW_u \approx \sigma(x) (W_{t_k} - W_s) + \sigma'(x) \sigma(x) \int_s^{t_k} \int_s^u dW_r dW_u.$$

The iterated Itô integral evaluates in closed form. With  $\Delta W := W_{t_k} - W_s$ ,

$$\int_s^{t_k} \int_s^u dW_r dW_u = \int_s^{t_k} (W_u - W_s) dW_u = \frac{1}{2} [(\Delta W)^2 - (t_k - s)] = \frac{1}{2} [(\Delta W)^2 - h],$$

where the  $-\frac{1}{2}h$  is exactly the Itô correction (the  $\int W dW = \frac{1}{2}(W^2 - t)$  identity from the martingale example). Writing  $\Delta W = \sqrt{h} Z_k$ , so  $(\Delta W)^2 - h = h(Z_k^2 - 1)$ , the one-step update becomes

$$\boxed{\hat{X}_{t_k} = \hat{X}_{t_{k-1}} + \mu h + \sigma \sqrt{h} Z_k + \frac{1}{2} \sigma \frac{\partial \sigma}{\partial x} (Z_k^2 - 1) h}$$

with all coefficients evaluated at  $(t_{k-1}, \hat{X}_{t_{k-1}})$ . The extra term  $\frac{1}{2} \sigma \partial_x \sigma (Z_k^2 - 1) h$  is the Milstein correction: it captures the curvature of the diffusion ( $\partial_x \sigma \neq 0$ ) over the step. Note it vanishes for *additive* noise ( $\sigma$  constant), where Euler and Milstein coincide — and indeed for additive noise Euler already has strong order 1.

**Remark 3.** Milstein needs the derivative  $\partial_x \sigma$  in closed form, and the multidimensional version requires iterated integrals  $\int \int dW^{(i)} dW^{(j)}$  (Lévy areas) for  $i \neq j$  that cannot be written in terms of the increments alone — this is why the slides note that Milstein “is not trivial to generalise to multi-dimensional SDEs.” For the Heston variance process  $dv_t = \kappa(\bar{v} - v_t) dt + \gamma \sqrt{v_t} dW_t$ ,  $\sigma(v) = \gamma \sqrt{v}$ ,  $\partial_v \sigma = \gamma/(2\sqrt{v})$ , so  $\sigma \partial_v \sigma = \gamma^2/2$  and the Milstein term is  $\frac{1}{4} \gamma^2 (Z^2 - 1) h$  — this is the standard full-truncation / Milstein scheme that mitigates (but does not eliminate) the negative-variance problem of plain Euler on Heston.

## 4 Strong vs. weak convergence

There are two error measures for a discretisation scheme with step  $h$ , depending on whether we care about the pathwise value or only a distributional functional.

**Strong order  $\beta$ .** Pathwise (root-mean /  $L^1$ ) accuracy:

$$\mathbb{E}[|\hat{X}_{Kh} - X_T|] \leq c h^\beta$$

for some constant  $c$  and all small  $h$ . This requires the simulated path and the true path (driven by the *same* Brownian motion) to stay close.

**Weak order  $\beta$ .** Distributional accuracy: for all sufficiently smooth test functions  $f$  (formally  $f \in C_P^{2\beta+2}$ , derivatives up to order  $2\beta + 2$  of polynomial growth),

$$|\mathbb{E}[f(\hat{X}_{Kh})] - \mathbb{E}[f(X_T)]| \leq c h^\beta.$$

Only the *laws* of  $\hat{X}_{Kh}$  and  $X_T$  need to match, not the paths.

## Orders.

Euler–Maruyama: strong  $\beta = \frac{1}{2}$ , weak  $\beta = 1$ ; Milstein: strong  $\beta = 1$ , weak  $\beta = 1$ .

Milstein’s payoff is the *strong* rate ( $\frac{1}{2} \rightarrow 1$ ); its weak rate equals Euler’s. So if all you want is a European price — an expectation  $\mathbb{E}[f(X_T)]$  — Euler already gives weak order 1 and Milstein buys you nothing distributionally. Milstein matters for *pathwise* accuracy: path-dependent payoffs (barriers, Asians, lookbacks) where the whole trajectory enters, and for any scheme where the same noise is reused (e.g. multilevel MC, which exploits strong convergence to couple coarse and fine levels).

**Remark 4** (The  $h^{1/3}$  reality check). *The weak-order-1 statement requires the test function  $f$  to be smooth. Option payoffs like  $(S_T - K)^+$  are not differentiable at the strike, so the theoretical  $\beta = 1$  is not guaranteed. Broadie–Kaya (2006) report empirical rates of  $O(h^{1/3})$  or worse for Euler on European prices under stochastic volatility. The slides’ heuristic: balance bias  $\sim h$  against MC noise  $\sim N^{-1/2}$  at total cost  $s \sim N/h$ . Eliminating  $N = sh$  gives error  $\sim h + (sh)^{-1/2}$ ; optimising over  $h$  (setting  $h \sim (sh)^{-1/2}$ , i.e.  $h \sim s^{-1/3}$ ) yields overall error  $O(s^{-1/3})$  in the computational budget  $s$ . This is much worse than the deterministic methods of this course and is the practical motivation for COS/FFT on models with a known characteristic function: they sidestep path discretisation entirely.*

## 5 Variance reduction

All four techniques attack the same target: shrink  $\text{Var}(\bar{X}_N) = \sigma^2/N$  by lowering the effective  $\sigma^2$ , keeping the estimator unbiased (or correcting the bias). A method giving variance  $\rho \sigma^2$  with  $\rho < 1$  delivers the accuracy of  $N/\rho$  plain samples at cost  $N$  — a speed-up of  $1/\rho$ .

### 5.1 Antithetic variates

**Construction (uniform case).** To estimate  $I = \mathbb{E}[f(U)]$ ,  $U \sim \mathcal{U}(0, 1)$ , pair each draw  $U_i$  with its reflection  $1 - U_i$  (also  $\mathcal{U}(0, 1)$ ):

$$\hat{I}_M = \frac{1}{M} \sum_{i=1}^M \frac{f(U_i) + f(1 - U_i)}{2}.$$

Since  $1 - U_i \sim \mathcal{U}(0, 1)$ , each summand has mean  $I$ , so  $\mathbb{E}[\hat{I}_M] = I$ : the estimator is unbiased.

**Variance.** For one pair,

$$\text{Var}\left(\frac{f(U) + f(1 - U)}{2}\right) = \frac{1}{4} \text{Var}(f(U)) + \frac{1}{4} \text{Var}(f(1 - U)) + \frac{1}{2} \text{Cov}(f(U), f(1 - U)).$$

Because  $U$  and  $1 - U$  have the same law,  $\text{Var}(f(U)) = \text{Var}(f(1 - U))$ , so this collapses to

$$\text{Var}\left(\frac{f(U) + f(1 - U)}{2}\right) = \frac{1}{2} \text{Var}(f(U)) + \frac{1}{2} \text{Cov}(f(U), f(1 - U)),$$

hence  $\text{Var}(\hat{I}_M) = \frac{1}{M} \left[ \frac{1}{2} \text{Var}(f(U)) + \frac{1}{2} \text{Cov}(f(U), f(1 - U)) \right]$ .

**When it helps.** Compare with the variance of the *plain* estimator using the same  $2M$  function evaluations,  $\frac{1}{2M} \text{Var}(f(U))$ . Antithetic wins iff

$$\text{Cov}(f(U), f(1 - U)) < 0.$$

This is guaranteed when  $f$  is monotone. The argument: if  $f$  is monotone increasing, then  $g(x) := -f(1-x)$  is also monotone increasing; by the FKG/correlation lemma (two comonotone functions of the same random variable have nonnegative covariance),  $\text{Cov}(f(U), -f(1-U)) \geq 0$ , i.e.  $\text{Cov}(f(U), f(1-U)) \leq 0$ . (The lemma: if  $f, g$  are both increasing,  $\text{Cov}(f(X), g(X)) \geq 0$ .) The Gaussian version is identical with the reflection  $U \mapsto -U$  (both  $\mathcal{N}(0, 1)$ ): along a path one runs the mirror path  $-Z_0, \dots, -Z_{N-1}$  and averages the two payoffs. For a vanilla call the discounted payoff is monotone in the driving normal, so antithetic always reduces variance there.

**Remark 5.** *Antithetic can fail (raise variance) when  $f$  is strongly non-monotone, e.g. a symmetric straddle or a payoff symmetric about the forward, where  $f(U)$  and  $f(1-U)$  become positively correlated. The monotonicity condition is the safe guarantee.*

## 5.2 Control variates: the optimal coefficient $c^*$

**Setup.** We want  $\mathbb{E}[X]$  (e.g. the arithmetic-Asian price). Suppose we can simulate, on the same paths, a correlated  $Y$  (e.g. the geometric-Asian payoff) whose mean  $\mathbb{E}[Y]$  is *known* in closed form. For any coefficient  $\theta$  define

$$Z_\theta = X + \theta(\mathbb{E}[Y] - Y).$$

Then  $\mathbb{E}[Z_\theta] = \mathbb{E}[X] + \theta(\mathbb{E}[Y] - \mathbb{E}[Y]) = \mathbb{E}[X]$  for every  $\theta$ : unbiased regardless of  $\theta$ . Only the variance depends on  $\theta$ :

$$\text{Var}(Z_\theta) = \text{Var}(X) - 2\theta \text{Cov}(X, Y) + \theta^2 \text{Var}(Y).$$

**Optimisation.** This is a quadratic in  $\theta$ , minimised by setting the derivative  $-2 \text{Cov}(X, Y) + 2\theta \text{Var}(Y) = 0$ :

$$\theta^* = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}$$

(Completing the square,  $\text{Var}(Z_\theta) = \text{Var}(Y)(\theta - \frac{\text{Cov}(X, Y)}{\text{Var}(Y)})^2 + \text{Var}(X) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)}$ , exposing both the minimiser and the minimum.) Note: with the slides' sign convention  $Z = X + \theta(\mathbb{E}Y - Y)$ , the optimum is  $\theta^* = + \text{Cov}(X, Y) / \text{Var}(Y)$ ; if instead one writes  $Z = X + c(Y - \mathbb{E}Y)$ , the same algebra gives  $c^* = - \text{Cov}(X, Y) / \text{Var}(Y)$  — the textbook " $c^* = - \text{Cov} / \text{Var}$ ." They are the same estimator; only the sign convention differs.

**Resulting variance reduction.** Substituting  $\theta^*$ ,

$$\text{Var}(Z_{\theta^*}) = \text{Var}(X) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)} = (1 - \rho_{XY}^2) \text{Var}(X), \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

The variance is reduced by the factor  $1 - \rho_{XY}^2$ . The reduction is dramatic exactly when  $X$  and  $Y$  are highly correlated: a control variate with  $|\rho| = 0.99$  cuts variance by  $\approx 50\times$ . The classic example: price the arithmetic-average Asian call (no formula) using the geometric-average Asian call (closed form for equally spaced fixings) as  $Y$ ; the two payoffs are nearly perfectly correlated along each path.

**Practical point.**  $\text{Cov}(X, Y)$  and  $\text{Var}(Y)$  are usually unknown and are *estimated from the same simulation* (a regression of  $X$  on  $Y$  gives  $\hat{\theta}^*$ ). Plugging in an estimated  $\hat{\theta}^*$  introduces a small  $O(1/N)$  bias, negligible against the  $O(N^{-1/2})$  statistical error; a clean fix is to estimate  $\theta^*$  on a pilot run and apply it on an independent run.

### 5.3 Importance sampling

**Idea and the likelihood ratio.** To estimate  $\mathbb{E}_p[f(X)] = \int f(x)p(x) dx$  when the "important" region (where  $f p$  is large) is rarely hit by  $p$  — the canonical case being deep out-of-the-money options, where the payoff is nonzero only on a low-probability set — sample instead from a proposal  $q$  and reweight:

$$\mathbb{E}_p[f(X)] = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_q\left[f(X) \frac{p(X)}{q(X)}\right].$$

The factor  $p(X)/q(X)$  is the *likelihood ratio* (Radon–Nikodym derivative  $dp/dq$ ); it corrects the bias from sampling under the "wrong" measure, leaving the estimator unbiased. The estimator is  $\frac{1}{N} \sum_i f(X_i) \frac{p(X_i)}{q(X_i)}$  with  $X_i \sim q$ .

**Gaussian drift shift.** For  $V = \mathbb{E}[f(X)]$  with  $X \sim \mathcal{N}(0, I_d)$ , the natural proposal is a mean shift  $X = Z + \theta$ ,  $Z \sim \mathcal{N}(0, I_d)$ . The likelihood ratio of  $\mathcal{N}(0, I)$  w.r.t.  $\mathcal{N}(\theta, I)$  evaluated at the sampled point is  $\exp(-\theta^\top Z - \frac{1}{2}\|\theta\|^2)$ , so

$$V = \mathbb{E}\left[f(Z + \theta) \exp(-\theta^\top Z - \frac{1}{2}\|\theta\|^2)\right].$$

(Scalar derivation: with  $f_0(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$  and  $g_\mu(z) = \frac{1}{\sqrt{2\pi}} e^{-(z-\mu)^2/2}$ , the ratio is  $f_0(z)/g_\mu(z) = \exp(-\frac{z^2}{2} + \frac{(z-\mu)^2}{2}) = \exp(-\mu z + \frac{1}{2}\mu^2)$ .) The shift  $\theta$  is chosen to move mass into the payoff region (e.g. for a call, shift the driving normal so that  $S_T > K$  becomes typical). Optimal  $\theta^*$  minimises the second moment  $M(\theta) = \mathbb{E}[f(Z + \theta)^2 \exp(-2\theta^\top Z - \|\theta\|^2)]$ , solving  $\nabla_\theta M(\theta^*) = 0$ .

**The double-edged sword.** A well-chosen  $q$  can reduce variance by orders of magnitude; a poorly chosen one (light tails relative to  $f p$ , so the ratio  $p/q$  blows up) gives an estimator with *infinite* variance — worse than plain MC. Importance sampling is the only one of the four techniques that can backfire catastrophically. Choosing  $q$  is "the art of importance sampling."

**Remark 6** (Girsanov / change of numeraire). *In continuous time the change of measure is the Girsanov theorem: shifting the drift of  $W$  by  $Y_s$  corresponds to the density process  $Z_t^{P \rightarrow Q} = \exp(\int_0^t Y_s dW_s^P - \frac{1}{2} \int_0^t Y_s^2 ds)$ , a martingale (under Novikov) with  $\mathbb{E}^P[Z_t] = 1$ . Change of numeraire is a special, financially-motivated change of measure: using the stock  $S_t$  as numeraire turns  $\mathbb{E}^Q[S_T \mathbf{1}_{\{S_T > K\}}]$  into  $e^{rT} S_0 Q^S(S_T > K) = e^{rT} S_0 N(d_1)$ , recovering the  $N(d_1)$  term of Black–Scholes by a measure change rather than by integration. Importance sampling does not require the proposal to come from a numeraire, but the two share the Radon–Nikodym machinery.*

## 6 Where this sits in the course

- **MC is the benchmark.** Every fast method in this course — COS (L6, L10), Carr–Madan FFT, finite differences (L3, L5), trees (L1) — is validated against a brute-force MC price with a CLT confidence interval. When the oral asks "how do you know your COS price is right," the honest answer is: it agrees with MC to within the MC standard error, and MC is unbiased by construction.
- **Rate comparison.** MC:  $O(N^{-1/2})$ , dimension-free. QMC:  $O((\log N)^d/N)$ , near- $O(N^{-1})$  in low effective dimension. Quadrature:  $O(N^{-k/d})$ , cursed in  $d$ . COS: *exponential* in the number of cosine terms for densities with known, smooth characteristic functions. This ordering is exactly why the course pivots to Fourier methods: for a 1-D European option under an affine model, COS reaches machine precision with a few hundred terms, where MC would need  $\sim 10^{12}$  paths.

- **When MC still wins.** High-dimensional baskets, path-dependent payoffs with no transform structure, and American/Bermudan via Longstaff–Schwartz (L9) — MC’s dimension-robustness keeps it indispensable. Variance reduction is what makes it competitive in those regimes.

## Honest gaps / what the slides do *not* prove

- **The slides do not derive Milstein from the Itô–Taylor expansion.** They state the scheme and the convergence orders but only *assert* that the extra term “accounts for the curvature of the diffusion.” The full derivation (Section 3.2 above) is mine, filling that gap; an examiner could reasonably ask for it.
- **The strong/weak orders are stated, not proved.** The slides give the definitions and quote  $\beta_{\text{Euler}} = (\frac{1}{2}, 1)$ ,  $\beta_{\text{Milstein}} = (1, 1)$  without proof. Proofs (Kloeden–Platen) are out of scope but worth knowing exist.
- **The optimal importance-sampling drift is left implicit.** The slides reduce to the first-order condition  $\nabla_{\theta} M(\theta^*) = 0$  but do not solve it in closed form (there is none in general; for a call one uses the asymptotically optimal shift placing the mode of the shifted integrand at the strike).
- **No explicit confidence-interval slide.** The CLT and standard error are on the slides, but the confidence-interval formula and the sample-variance estimator  $s_N^2$  (Section 1.3) are standard background I have added.
- **QMC’s effective-dimension story is absent.** The slides give the Koksma–Hlawka bound and the  $(\log N)^d/N$  rate but do not discuss why QMC works in practice for high-dimensional path simulation (Brownian bridge / PCA path construction to lower effective dimension).